

EEG Feature extraction and pattern classification based on Motor Imagery in Brain-computer Interface

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Abstract—Accurate classification of left and right hand motor imagery of EEG is an important issue in brain-computer interface (BCI). Here, discrete wavelet transform was firstly applied to extract the features of left and right hand motor imagery in EEG. Secondly, Fisher Linear Discriminant Analysis was used with two different threshold calculation methods and obtained good misclassification rate. We also used Support Vector Machine to compare the performance with Fisher Linear Discriminant Analysis. The final classification results showed that false classification rate by Support Vector Machine was the lowest and gained a ideal classification results.

Keywords—Brain-computer interface; motor imagery; feature extraction; pattern classification

I. INTRODUCTION

Brain-computer interface (BCI) is a communication system by which a person can send messages without any use of peripheral nerves and muscles [1-4]. The technology holds great promise for people who can't use their arms or hands normally because they have had the damaged region, such as amyotrophic lateral sclerosis (ALS), spinal cord injury, brainstem stroke, or quadriplegic patients. BCI system, by extracting signals directly from the brain, might help to restore abilities to patients who have lost sensory or motor function because of their disabilities. In fact, BCI is used as a surrogate for the people who are disabled, in the case of a neuromotor prosthesis, acts to interpret brain signals and drive the appropriate effector (e.g., muscles or a robotic arm)[5,6]. So, it has great social value and application prospect. As an interdisciplinary technology, BCI has become a research hotspot of brain cognition research, rehabilitation engineering, biomedical engineering, automatic human-machine control, and so on.

A BCI has three basic components: the signal acquisition component, which records brain signals; the signal processing component, which includes the software that extracts the features of the brain signals that are used for the BCI and the translation algorithm that translates the extracted features into device commands; the output device component that implements the commands.

A number of groups around the world are developing BCI systems, in which surface electroencephalography (EEG) is

used to record the signals that control cursor movements on a computer screen [1, 6, 7]. The cognitive task most commonly used in BCI studies is motor imagery, as it produces changes in EEG that occur naturally in movement planning and are relatively straightforward to detect. The signals generated in the motor cortex can be recorded from electrodes over central head regions and the research has produced encouraging results. With the operant conditioning approach, the intention is to train subjects to control the cursor automatically [7].

Feature extraction and pattern classification of EEG is a difficult problem during the research of BCI. In this paper, oriented to the left and right hand motor imagery based BCI, a new method for the recognition of electroencephalography (EEG) was proposed, which combined the discrete wavelet transform (DWT), Fisher Linear Discriminant Analysis (FLDA) and Support Vector Machine (SVM). Classification results obtained by FLDA and SVM based on the feature set by DWT are effective, especially good for SVM.

II. METHODS

In this paper, we used data collected during online BCI experiments, and analyzed them offline as described later in this section.

A. Description of data set

Data set was provided by Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology Graz [8].

This dataset was recorded from a normal subject (female, 25y) during a feedback session. The subject sat in a relaxing chair with armrests. The task was to control a feedback bar by means of imagery left or right hand movements. The order of left and right cues was random. The electrode positions and timing scheme was showed in Fig.1.

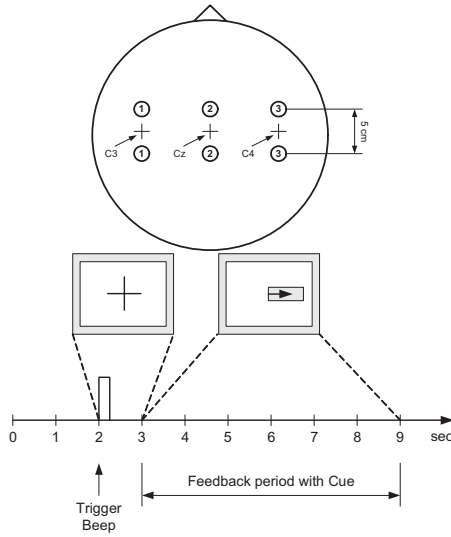


Figure 1. Electrode positions (top) and timing scheme (bottom) for recording the Graz BCI data set

The experiment consists of 7 runs with 40 trials each. All runs were conducted on the same day with several minutes break in between. Given are 280 trials of 9s length. The first 2s was quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial, the trigger channel (#4) went from low to high, and a cross “+” was displayed for 1s; then at $t=3s$, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of the cue. The recording was made using a G.tec amplifier and a Ag/AgCl electrodes. Three bipolar EEG channels (anterior ‘+’, posterior ‘-’) were measured over C3, Cz and C4. The EEG was sampled with 128Hz; it was filtered between 0.5 and 30Hz.

The trials for training and testing were randomly selected. This should prevent any systematic effect due to the feedback.

B. Feature extraction

Because channel Cz showed its independence of the motor imagery, only channel C3 and C4 were used for feature extraction. The duration of each experiment was 9 seconds, but the part which played a role for the task of classification may be just one period. In order to find a suitable time window, the average power of different imagine tasks of C3 and C4 channel was computed firstly, so as to provide effective reference of the next step.

The formula for calculating average power is:

$$P_{(j)} = \frac{1}{N} \sum_{i=1}^N x_{f(i,j)}^2 \quad (1)$$

$x_{f(i,j)}^2$ is the j EEG data of the i time experiment, $P_{(j)}$ is the average power of the j EEG data corresponding to the N time experiments.

We could calculate the average power PLC3 and PLC4 of

C3,C4 when one imagined left-hand movement corresponding to 0~9 seconds, and the average power PRC3 and PRC4 of C3,C4 when one imagined right-hand movement corresponding to 0~9 seconds. As shown in Fig. 2 and Fig. 3.

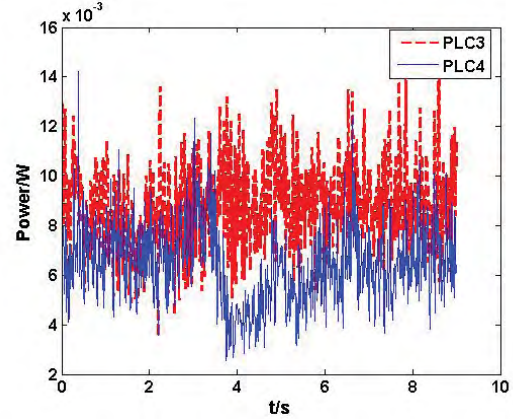


Figure 2. The relationship between the time and the average power of the C3 electrode and C4 electrode for left-hand imagery movement

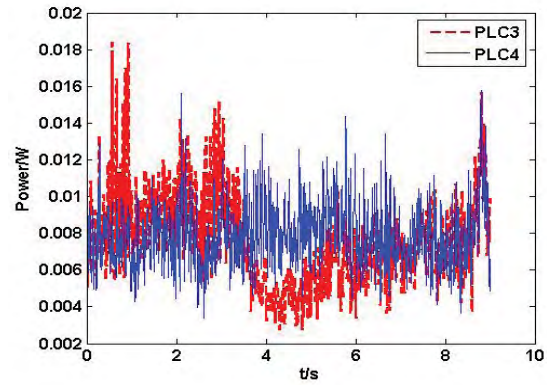


Figure 3. The relationship between the time and the average power of the C3 electrode and C4 electrode for right-hand imagery movement

From Fig.2 and Fig.3, there were significant differences between the energy of left-right hand imagery movements. The average power obtained by C3-channel was greater than C4-channel when imagined left hand movement; the average power obtained by C3-channel was less than C4-channel when imagined right hand movement. It was in line with ERD/ERS features of motor imagery. Event-related desynchronization potentials generated in contralateral and event-related synchronization potentials generated in ipsilateral of the brain when one has unilateral limb movement or imagined movement.

Thus, we could get the average power at 3.5~8 seconds. Taking into account the similarities between the wavelet function and the signal waveform to be analyzed, this paper selected Daubechies (dbN) wavelet system. Experimental results showed that the reconstructed signal could basically reflect the characteristics of the original signal when used db5. So, we had a 6-layer decomposition of average power for left-hand imagery movement with db5 wavelet function in the time window, that is

$$f(t) = A_6 + D_6 + \dots + D_1 \quad (2)$$

Decomposition results were shown in Fig. 4 and Fig. 5.

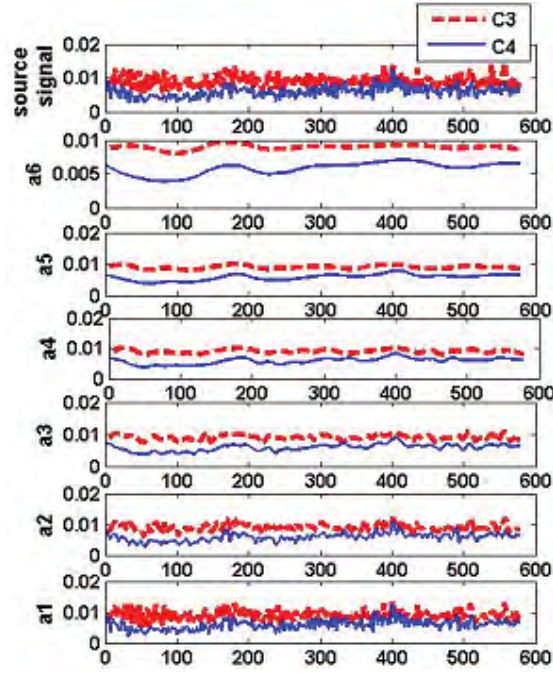


Figure 4. The wavelet decomposition of the C3 electrode and C4 electrode for left-hand imagery movement

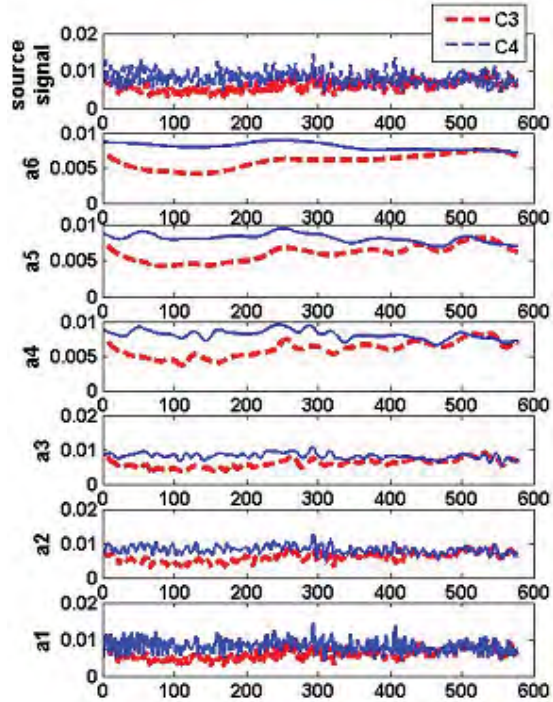


Figure 5. The wavelet decomposition of the C3 electrode and C4 electrode for right-hand imagery movement

Here, L3A6 and L4A6 were the approximation signals at the scale 6 of C3 and C4 when imagined left-hand movement, R3A6 and R4A6 were the approximation signals at the scale 6 of C3 and C4 when imagined right-hand movement. The results showed that there was significant difference between left-hand and right-hand imagery movement concerning the signal of approximation coefficient A6 on the sixth level. The value of L3A6 was bigger than L4A6 when imagined left-hand movement, while the value of R3A6 was smaller than R4A6 when imagined right-hand movement. Supposing LA6 and RA6 were the difference of the signal of approximation coefficient A6 on the sixth level of C3 and C4, respectively.

$$LA6 = L3A6 - L4A6 \quad (3)$$

$$RA6 = R3A6 - R4A6 \quad (4)$$

The curve of LA6 and RA6 was shown in Fig.6. Obviously, LA6 and RA6 can be selected as the features for distinguish the left-right hands imagery movement.

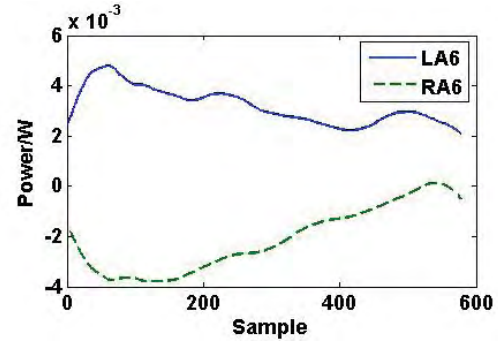


Figure 6. The feature signals of the left-right hands imagery movement

C. Classification

In order to demonstrate the effectiveness of the proposed features in BCI application, the FLDA classification method and SVM method were used in this paper. For the sake of self-containment, this section briefly introduced the FLDA and SVM.

1) FLDA classifier

The goal in Fisher's linear discriminant analysis (FLDA) was to compute a discriminate vector that separates two or more classes as well as possible. Here we considered the two-class case. We were given a set of input vectors $x_i \in R^D$, $i \in \{1 \dots N\}$ and the corresponding class-labels $y_i = \{-1, 1\}$. Denoting by N_1 the number of training examples for which $y_i = 1$, by C_1 the set of indices i for which $y_i = 1$, and using analogous definitions for N_2, C_2 , the objective function for computing a discriminate vector $w \in R^D$ is

$$J(w) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} \quad (5)$$

Where

$$\mu_k = \frac{1}{N_k} \sum_{i \in C_k} w^T x_i, \quad (6)$$

$$\sigma_k^2 = \sum_{i \in C_k} (w^T x_i - \mu_k)^2 \quad (7)$$

This means that one was searching for discriminate vectors that result in a large distance between the projected means and small variance around the projected means (small within-class variance). To compute directly the optimal discriminate vector for a training data set, matrix equations for the quantities $(\mu_1 - \mu_2)^2$ and $\sigma_1^2 + \sigma_2^2$ can be used. We first define the class means m_k :

$$m_k = \frac{1}{N_k} \sum_{i \in C_k} x_i \quad (8)$$

Now we can define the between-class scatter matrix S_B and the within-class scatter matrix S_W .

$$S_B = (m_1 - m_2)(m_1 - m_2)^T \quad (9)$$

$$S_W = \sum_{k=1}^2 \sum_{i \in C_k} (x_i - m_k)(x_i - m_k)^T \quad (10)$$

With the help of these two matrices, the objective function for FLDA can be written as a Rayleigh quotient

$$J(w) = \frac{w^T S_B w}{w^T S_W w} \quad (11)$$

By computing the derivative of J and setting it to zero, one can show that the optimal solution for w satisfies the following equation:

$$w \propto S_W^{-1} (m_1 - m_2) \quad (12)$$

The classification was conducted as follows:

$$x \in \begin{cases} \text{class } 1 & , \text{ if } y > 0 \\ \text{class } 2 & , \text{ if } y < 0 \end{cases} \quad (13)$$

2) SVM classifier

SVM was a classification algorithm based on statistical learning theory. The invention of SVM was driven by underlying statistical learning theory, i.e., following the principle of structural risk minimization that was rooted in VC dimension theory, which makes its derivation even more profound[6,7]. The SVM has been a topic of extensive research with wide applications in machine learning and engineering. The output of a binary SVM classifier can be computed by the following expression:

$$f(x) = \text{sgn} \left(\sum_{i=1}^n \alpha_i^* y_i K(x_i, x) + b^* \right) \quad (14)$$

where $\{x_i, y_i\}_{i=1}^N$ were training samples with input vector $x_i \in R^d$; class labels $y_i \in \{-1, 1\}$, $\alpha_i \geq 0$ were Lagrangian multipliers obtained by solving a quadratic optimization problem; b was the bias, and $k\{x_i, x_j\}$ was called kernel function in SVM. There were two typical kernel functions: linear kernel and RBF kernel:

$$K(x_i, x_j) = \begin{cases} x_i^T x_j \\ \exp\left(-\frac{1}{2\sigma^2} \|x - x_i\|^2\right) \end{cases} \quad (15)$$

where σ was the kernel parameter. It was fair to note that the parameters C and σ should be tuned properly for accurate prediction of unseen samples. In this study, they were selected using LIBSVM by 5-fold cross-validation over the following grid of (C, σ) : $[2^{-7} \dots 2^7] \times [2^{-10} \dots 2^3]$.

The protruding characteristics of SVM lies in its elegant mechanism of handling nonlinear function classes[9,10], i.e., nonlinear information processing is carried out by means of linear techniques in an implicit high-dimensional feature space mapped by a nonlinear transformation $\phi(x_i)$ from original input space. Although the analytical expressions of $\phi(x_i)$ is unknown, only the inner product operations $\phi(x_i)^T \phi(x_j)$ are involved, the kernel functions can be used to substitute the inner product operations according to the Mercer theorem. Vapnik's theory [10] showed that the SVM solution was found by minimizing both the error on the training set (empirical risk) and the complexity of the hypothesis space, expressed in terms of VC dimension. In this sense, the decision function found by SVM is a tradeoff between learning error and model complexity.

III. RESULTS AND DISCUSSION

Taking into account the different number of samples of experiments may have an impact on classification results, the raw data were carried out on grouping at the beginning of the experiment. The first group consisted of 70 training samples and 70 test samples, recorded as Data1. The second group consisted of 140 training samples and 70 test samples, denoted by Data2.

In this paper, two different methods of threshold calculating were used in achieving Fisher linear discriminant analysis algorithm. One threshold value was the mid-point of class center projection of two types. The other threshold was the weighted average based on the class frequency of two class centers. So as to facilitate the discussion below, here denoted FLDA1 and FLDA2 respectively. Classification results were shown in Fig.7 and Fig.8.

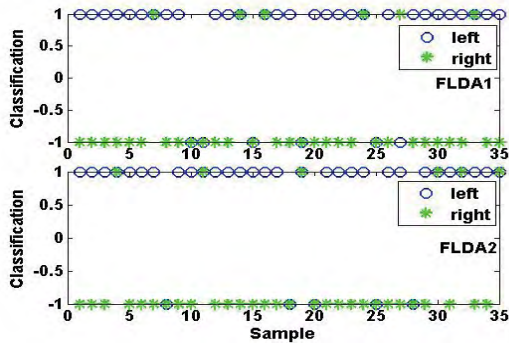


Figure 7. The classification result based on FLDA method and using Data1

○ denote imagined left hand movement (category marked was 1), * denote imagined right hand movement (category marked was -1). It can be seen from the Fig.7 that in FLDA1 classification, at the 10th point, ○ was assigned to the class -1, while the ○ belonged to category 1 originally; at the 7th point, * was assigned to the class 1, while * belonged to category -1 originally. Therefore, there have been misclassified. Likewise the meaning of the subsequent figure.

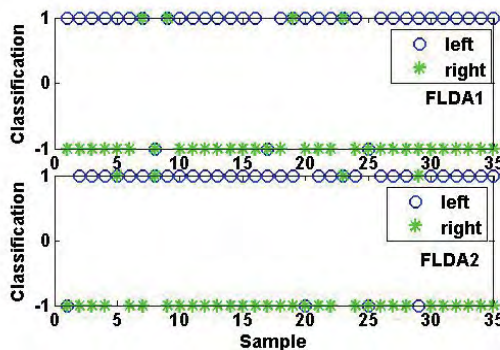


Figure 8. The classification result based on FLDA method and using Data2

The parameter selection of kernel function for SVM have experiencing method, trying method, as well as through a number of optimization methods, where the method described in Section II was used. After determining these parameters, we

can identify unknown samples by the classification algorithm. Ref and Linear kernel function were used in SVM. Classification results were shown in Fig. 9 and Fig.10.

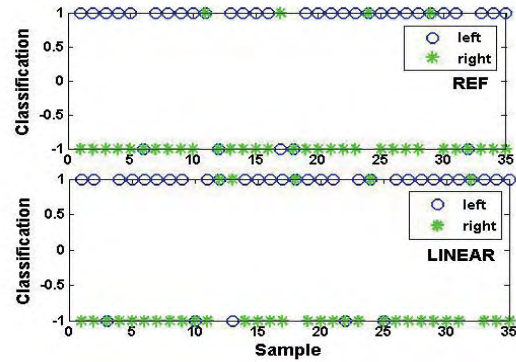


Figure 9. The classification result based on SVM method and using Data1

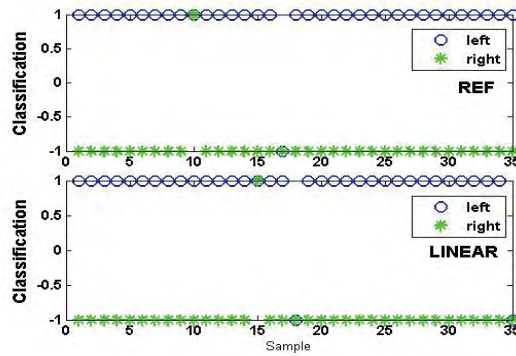


Figure 10. The classification result based on SVM method and using Data2

In this experiment, Misdiscriminant rate (MR) is the main criteria for measure of classifier performance.

$$MR = \frac{\text{classified incorrectly samples}}{\text{sample population}} \times 100\% \quad (16)$$

TABLE I. CLASSIFICATION RESULT OF DIFFERENT METHODS

Discriminate criterion		FLDA		SVM	
		FLDA1	FLDA2	RBF	LINEAR
MR	Data1	17.65	16.18	14.58	14.65
%	Data2	10.29	11.24	3.94	4.28

Experimental results were shown in Table1. In the case of using Data1(or Data2) data set, when we used Fisher linear discriminant analysis method with different threshold calculated, the results obtained were almost the same, this is because the number of left-right hand motor imagery of raw EEG data was quite close, when we used SVM with different kernel function method for classification, the two classification results were quite close, which indicated support vector machine is not very sensitive for the choice of kernel function. The results obtained from Data2 were superior to the results

obtained from data1 for both algorithms. It is indicated that recognition rate increased with the the training set increased, which is more evident for SVM classifier.

The above experimental results showed that when identifying the different mental tasks from EEG signals, the LDA classifier, especially SVM classifier are very effective based on DWT feature extraction.

The experimental results indicated the rational of feature extraction and classification algorithm, which also accumulated experience and laid the foundation for the following study of online brain-computer interface system.

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